GABRIEL POPA

Senior Software Engineer – AI/ML

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ABOUT ME

Senior AI/ML Engineer with 10+ years of experience building **production-grade machine learning systems**, **data pipelines**, and **model-powered backend platforms** across business applications, analytics workflows, and operational systems. Strongest work centers on turning raw business data into reliable **training and inference pipelines**, deploying models into stable production environments, improving **feature engineering** and **model monitoring** practices, and supporting end-to-end **MLOps** workflows from experimentation through serving and continuous improvement. Hands-on across **Python**, **SQL**, **machine learning pipelines**, **batch and real-time inference**, **model deployment**, **feature stores**, **workflow orchestration**, **Docker**, **cloud services**, **monitoring**, and safe rollout practices for high-availability AI-enabled systems.

SKILLS

- **AI / Machine Learning**: Supervised learning, classification, regression, feature engineering, model evaluation, hyperparameter tuning, training pipelines, batch inference, real-time inference, model serving, ML experimentation, model validation, drift monitoring, model performance tracking
- **Programming:** Python, SQL, Bash, REST APIs, backend service development, data transformation, ETL, integration workflows
- **ML / Data Tooling:** Pandas, NumPy, scikit-learn, XGBoost, TensorFlow, PyTorch, Jupyter, MLflow, Airflow, dbt, feature engineering frameworks
- **Data Platforms:** PostgreSQL, MySQL, SQL Server, Redis, MongoDB, data warehousing, dataset preparation, schema design, query tuning, data quality validation
- **Cloud & MLOps:** AWS, Azure, GCP, Docker, Kubernetes, CI/CD, model packaging, artifact versioning, deployment automation, workflow orchestration, environment reproducibility
- **Observability & Quality**: Logging, metrics, alerting, experiment tracking, unit testing, integration testing, data validation, model monitoring, performance benchmarking, production incident support

PROFESSIONAL EXPERIENCE

Senior Software Engineer

Soft Build | Bucharest, Romania *(Jan 2024 – May 2026)*

- Led delivery of **production AI/ML services** that exposed **model inference APIs** and automated scoring jobs for operational decision workflows.
- Designed and maintained end-to-end **ML pipelines** using **Airflow**, **Pandas**, **NumPy**, and **SQL** for ingestion, transformation, feature preparation, training, and deployment.
- Built reusable **feature engineering** workflows with **Pandas**, **NumPy**, and **dbt**, improving dataset consistency by **24%** across model training runs.
- Trained and evaluated production models using **scikit-learn**, **XGBoost**, and **TensorFlow**, improving prediction quality and supporting multiple business use cases.
- Implemented **model deployment** patterns for containerized services, reducing deployment friction by **21%** and improving release confidence.
- Developed **batch inference** pipelines and real-time scoring endpoints that reduced prediction turnaround time by **28%** compared with older asynchronous processing paths.
- Improved experiment traceability and retraining reproducibility through **MLflow**-based tracking, artifact versioning, and model registry practices, reducing failed retraining cycles by **19%**.
- Added validation checks for schema drift, missing values, and feature anomalies before training and serving, improving pipeline reliability by **23%**.
- Optimized orchestration and transformation jobs across **Airflow**, **dbt**, **Pandas**, and **NumPy**, reducing pipeline execution time by **26%**.

- Built monitoring for model latency, failure rates, and data freshness, helping reduce model-related production incidents by **17%**.
- Partnered with product and backend teams to operationalize ML outputs into stable user-facing and internal workflows.
- Used **Jupyter** for controlled experimentation and feature analysis before promoting validated logic into production pipelines and services.
- Supported **MLOps** practices across CI/CD, containerization, automated validation, and rollback-safe release procedures for AI-enabled systems.
- Helped define stronger internal standards for **model lifecycle management**, improving handoff quality between experimentation and production engineering teams.
- Contributed to more maintainable **production inference systems** by aligning training, registry, deployment, and monitoring workflows under one operating model.

Software Engineer

Next Apps | Ghent, Belgium *(Nov2020 – Nov 2023)*

- Built and maintained **machine learning data pipelines** that collected, normalized, and transformed multi-source application data for training and analytical workflows.
- Developed **feature engineering** jobs for behavioral, transactional, and operational datasets, improving model signal quality and increasing downstream model accuracy by **14%**.
- Implemented **training pipelines** for classification and regression use cases, covering dataset creation, split logic, evaluation reporting, and artifact packaging.
- Supported deployment of **model inference services** into cloud-hosted environments, improving availability and reducing manual release effort by **22%**.
- Created scheduled **batch scoring** workflows that delivered predictions to business systems for segmentation, prioritization, and forecasting use cases.
- Built internal tooling for experiment comparison and model result review, making it easier for teams to select stronger candidate models and reduce evaluation time by **18%**.
- Improved **data quality controls** with validation layers that caught incomplete and inconsistent records before training and serving stages.
- Worked on **MLOps** automation for packaging models, promoting artifacts between environments, and validating deployment readiness before release.
- Collaborated with application engineers to integrate **prediction outputs** into user-facing and internal workflows without disrupting existing business processes.
- Tuned pipeline SQL and transformation logic to improve training-data generation speed by **27%** on large datasets.
- Added service-level monitoring and alerting for model endpoints, helping reduce production recovery time by **16%** during inference failures.
- Contributed to documentation and shared engineering practices around **model deployment**, reproducibility, and safe operational support for ML-backed systems.
- Helped evolve backend architecture so **ML components** could be deployed and maintained more consistently alongside conventional application services.

Software Developer

Datamart | Stockholm, Sweden *(Feb 2016 – Sep 2020)*

- Built backend-oriented **data processing pipelines** that prepared structured business data for analytics, rule-based scoring, and early-stage machine learning workflows.
- Developed reusable **ETL** jobs for ingestion, cleansing, normalization, and aggregation across relational and semi-structured data sources.
- Created feature preparation scripts and dataset assembly logic used by analysts and data scientists for experimentation and predictive modeling.
- Improved dataset reliability through validation checks and transformation controls, reducing training-data defects by **20%**.
- Worked on **model-supporting backend services** that exposed processed data and scoring outputs through internal APIs and reporting systems.
- Automated recurring data preparation flows, reducing manual analyst effort by **31%** and enabling more consistent retraining inputs.

- Optimized SQL-heavy transformations and batch jobs to improve pipeline throughput by **25%** across large reporting and prediction-support workloads.
- Assisted with deployment of early **machine learning models** into business workflows by packaging outputs into operational systems and scheduled delivery paths.
- Collaborated with product and analytics stakeholders to define usable features from raw business data, strengthening **feature engineering** quality for forecasting and segmentation tasks.
- Contributed to monitoring and troubleshooting of long-running data jobs, improving operational stability and reducing failed runs by **15%**.
- Helped bridge analytics and engineering work by turning notebook-style experiments into more repeatable, production-oriented processing workflows.

Junior Software Developer

Berg Software | Timișoara, Romania *(Sep2014 – Mar 2016)*

- Supported development of backend and data-processing components for internal business applications with a strong focus on **Python**, **SQL**, and structured data handling.
- Built scripts for extraction, transformation, and loading of operational data used in reporting and early predictive analysis.
- Assisted senior engineers with preparation of model-ready datasets, including feature cleanup, categorical mapping, and missing-value handling.
- Improved processing reliability by adding validation and logging around ingestion and transformation steps, reducing avoidable job failures by **12%**.
- Wrote maintainable utility services and batch jobs that helped standardize repeated data preparation tasks across projects.
- Helped optimize database queries and intermediate transformations, improving job completion speed by **18%** on recurring workflows.
- Participated in testing and debugging of backend processes used for analytics and data-driven product features.
- Worked with teammates to document pipeline logic, transformation assumptions, and support procedures for production data jobs.
- Built early hands-on experience in **feature engineering**, **pipeline reliability**, and production support for data-intensive software systems.

EDUCATION

Bachelor's Degree in Computer Science

University of Bucharest | (*2010 – 2014)*

- Focused on practical software engineering fundamentals that translated directly into building reliable web systems and data-driven applications.